Double Categories

Lifting Programs to Specifications

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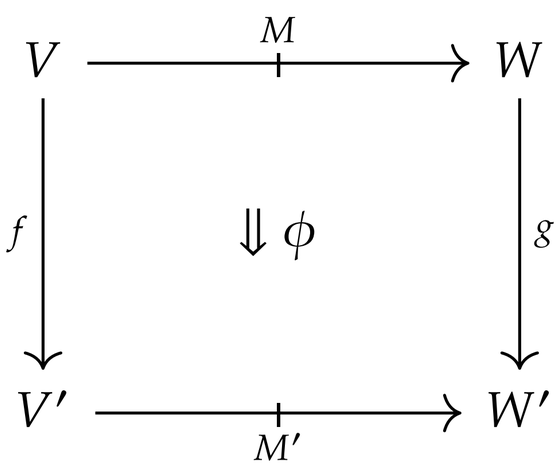
Introduction

A **double category** has both: *tight* arrows (functions), *loose* arrows (relations), and *squares* relating them. We use these to express how to **lift** programs (or functions) to specifications (or relations) of their behavior.

This is a reference for effectful programs as arrows Power and Robinson (1997).

Double categories

Definition 1 A double category has objects, tight arrows $f : V \to V'$, loose arrows $M : V \nrightarrow W$ and squares



with loose composition associative and unital up to coherent isomorphism.

Example

Example 1 $\mathbb{R}$el has sets, functions and relations $R \subseteq A \times B$. A square from R to R' along f, g exists when

$$\forall (x, y) \in R.\ (f(x), g(y)) \in R'.$$

Results

- First.
- Second.

References

Power, John, and Edmund Robinson. 1997. "Premonoidal Categories and Notions of Computation." *Mathematical Structures in Computer Science* 7 (5): 453–68.